TASK 3 COMPLETION REPORT

Real-Time GNC Loop Under Emulation

EduRankAI / VESPER Project | Corrected Cortex-M4 + FreeRTOS Timing Build

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Project: Real-Time GNC Loop Under Emulation

1. Objective

Task 3 focused on building and checking the corrected real-time GNC timing test under Renode emulation, inspecting task-period and deadline counters, checking the SysTick configuration, and collecting reproducible evidence for the four scheduled task loops: GNC, sensing, communications, and telemetry.

2. Build and Configuration

The corrected source file phase3_timing_hr_deadline_corrected.c and the generated executable phase3_timing_hr_deadline_corrected_test.elf were verified in the c_port directory. The Renode script phase3_timing_hr_deadline_corrected_test.resc was also present. The screenshot of the directory listing records the source, ELF, and script sizes as 17,251 bytes, 61,832 bytes, and 271 bytes respectively.

The test uses a Cortex-M4 STM32F4 Discovery emulation target, with Renode configured for 125 MIPS. The FreeRTOS configuration specifies configCPU_CLOCK_HZ = 72,000,000 and configTICK_RATE_HZ = 1,000. The inspected SysTick LOAD value is 0x0001193F (71,999), which is consistent with a 1 ms tick when the SysTick clock source is 72 MHz.

3. Task 3 Results

3.1 Nominal task periods and RTOS period observations

Task	Configured/nominal period	Observed RTOS period (min/max)
GNC	10 ms	10 / 10 ticks
Sensing	20 ms	20 / 20 ticks
Communications	50 ms	50 / 50 ticks
Telemetry	100 ms	100 / 100 ticks

3.2 Activation-period violation counters

Task / counter address	Observed value
GNC — 0x20000014	0
Sensing — 0x20000048	0
Communications — 0x2000007C	0
Telemetry — 0x200000B0	0

3.3 Deadline-miss counters

Task / counter address	Observed value
GNC — 0x20000018	0
Sensing — 0x2000004C	0
Communications — 0x20000080	0
Telemetry — 0x200000B4	0

3.4 High-resolution Timer2 observations (raw counts)

Timer2 results are reported as raw counter counts because the Timer2 input frequency was not independently verified. Therefore, these values are not converted to milliseconds or microseconds.

Task	Min period counts	Max period counts	Min execution counts	Max execution counts	Peak-to-peak period variation
GNC	99,255	100,027	90	94	772
Sensing	199,266	200,027	0	21	761
Communications	499,208	500,070	7	12	862
Telemetry	999,281	1,000,372	6	10	707

4. Interpretation of Results

- The observed RTOS task periods match the intended 10 ms, 20 ms, 50 ms, and 100 ms schedule at the 1 ms tick resolution.
- All four activation-period violation counters and all four deadline-miss counters read zero in the captured run.
- Timer2 period counts show bounded variation during the observed run. The peak-to-peak variations are 772 counts for GNC, 761 for sensing, 862 for communications, and 707 for telemetry.
- The RTOS period measurements are quantized to 1 ms ticks. A displayed execution count of zero in the sensing measurement does not mean that the task consumed no CPU time.
- The execution-count intervals include the measurement and bookkeeping instrumentation and should not be interpreted as pure application-workload WCET.

5. Work Completed

- Reviewed the corrected timing source and confirmed that the corresponding ELF and Renode .resc launch script exist in the project directory.
- Inspected all four activation-period violation counters and recorded zero for each task.
- Inspected all four deadline-miss counters and recorded zero for each task.
- Read the SysTick LOAD and current-value registers and recorded the configuration evidence.
- Inspected task period measurements for GNC, sensing, communications, and telemetry.
- Collected high-resolution Timer2 period and execution count readings for each of the four tasks.
- Organized the supplied screenshots into a figure sequence with explanatory captions for the report.

6. Evidence Figures

The following figures are the screenshots supplied for this report. Each caption describes what is visible and how it supports the Task 3 verification record.

```
0x00000000
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000014
0x00000000
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000048
0x00000000
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x2000007C
0x00000000
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000B0
0x00000000
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 1. Activation-period violation counters for the four FreeRTOS tasks. The displayed values are zero at the captured observation point, indicating that no activation-period violations had been recorded in this run.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000018
0x00000000
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x2000004C
0x00000000
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000080
0x00000000
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000B4
0x00000000
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 2. Deadline-miss counters for GNC, sensing, communications, and telemetry. All four displayed counters are zero at the captured observation point; this is evidence for the observed run, not a proof of a hard real-time guarantee.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0xE000E014
0x0001193F
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0xE000E018
0x0000A615
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 3. SysTick register inspection. LOAD at 0xE000E014 is 0x0001193F (71,999), consistent with a 1 kHz tick when driven by a 72 MHz SysTick clock. The current-value register is also shown for configuration evidence.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000004
0x0000000A
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000008
0x0000000A
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000038
0x00000014
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x2000003C
0x00000014
```

Figure 4. GNC and sensing task period measurements. The observed period counters are consistent with nominal periods of 10 ms for GNC and 20 ms for sensing at a 1 ms RTOS tick.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x2000006C
0x00000032
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000070
0x00000032
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000A0
0x00000064
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000A4
0x00000064
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 5. Communications and telemetry task period measurements. The observed period counters are consistent with nominal periods of 50 ms and 100 ms respectively.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000020
0x000183B7
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000024
0x000186BB
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000028
0x0000005A
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x2000002C
0x00000068
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000030
0x00000304
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 6. GNC task high-resolution timer measurements. Raw Timer2 period and execution counts are shown; the observed period-count range is 99,255–100,027, execution counts are 90–94, and peak-to-peak period variation is 772 counts.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000054
0x00030A62
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000058
0x00030D5B
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x2000005C
0x00000000
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000060
0x00000015
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000064
0x000002F9
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 7. Sensing task high-resolution timer measurements. Raw Timer2 period and execution counts are shown; the observed period-count range is 199,266–200,027, execution counts are 0–21, and peak-to-peak period variation is 761 counts.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000088
0x00079E08
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x2000008C
0x0007A16E
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000090
0x00000007
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000094
0x00000016
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x20000098
0x00000366
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 8. Communications task high-resolution timer measurements. Raw Timer2 period and execution counts are shown; the observed period-count range is 499,208–500,070, execution counts are 7–12, and peak-to-peak period variation is 862 counts.

```
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000BC
0x000F3F71
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000C0
0x000F4234
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000C4
0x00000006
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000C8
0x0000000A
(GNC_Phase3_Deadline_Corrected_Test)
(GNC_Phase3_Deadline_Corrected_Test) sysbus ReadDoubleWord 0x200000CC
0x000002C3
(GNC_Phase3_Deadline_Corrected_Test)
```

Figure 9. Telemetry task high-resolution timer measurements. Raw Timer2 period and execution counts are shown; the observed period-count range is 999,281–1,000,372, execution counts are 6–10, and peak-to-peak period variation is 707 counts.

```
C:\Users\DELL\Week4_GNC_Renode\c_port>cd /d C:\Users\DELL\Week4_GNC_Renode\c_port

C:\Users\DELL\Week4_GNC_Renode\c_port>dir phase3_timing_hr_deadline_corrected.c
 Volume in drive C is OS
 Volume Serial Number is AAD0-2DCC

 Directory of C:\Users\DELL\Week4_GNC_Renode\c_port

24-10-2026  21:17            17,251 phase3_timing_hr_deadline_corrected.c
               1 File(s)         17,251 bytes
               0 Dir(s)  15,310,938,112 bytes free

C:\Users\DELL\Week4_GNC_Renode\c_port>dir phase3_timing_hr_deadline_corrected_test.elf
 Volume in drive C is OS
 Volume Serial Number is AAD0-2DCC

 Directory of C:\Users\DELL\Week4_GNC_Renode\c_port

24-10-2026  21:23            61,832 phase3_timing_hr_deadline_corrected_test.elf
               1 File(s)         61,832 bytes
               0 Dir(s)  15,310,938,112 bytes free

C:\Users\DELL\Week4_GNC_Renode\c_port>dir phase3_timing_hr_deadline_corrected_test.resc
 Volume in drive C is OS
 Volume Serial Number is AAD0-2DCC

 Directory of C:\Users\DELL\Week4_GNC_Renode\c_port

24-10-2026  21:24               271 phase3_timing_hr_deadline_corrected_test.resc
               1 File(s)            271 bytes
               0 Dir(s)  15,310,938,112 bytes free

C:\Users\DELL\Week4_GNC_Renode\c_port>
```

Figure 10. Windows directory listing confirming the corrected C source, generated ELF executable, and Renode script are present in the project build directory. The displayed file sizes are 17,251 bytes for the C source, 61,832 bytes for the ELF, and 271 bytes for the .resc script.

7. Detailed Task Completion Record

This section records the work performed for Task 3, the checks made against the running emulation, the evidence collected, and the meaning of the observed results. It is intended to make the completion process traceable from the build artifacts through the runtime measurements.

7.1 Source, executable, and launch-script verification

The corrected C source file, phase3_timing_hr_deadline_corrected.c, was confirmed in the c_port directory. The corresponding phase3_timing_hr_deadline_corrected_test.elf executable and phase3_timing_hr_deadline_corrected_test.resc Renode script were also confirmed by the Windows directory listing. This establishes that the source, executable used for the test, and script used to load the target were present together in the project build folder. The directory screenshot is retained as build evidence.

7.2 Runtime activation-period checks

The activation-period violation counter for each task was read from emulated memory using Renode's sysbus ReadDoubleWord command. The recorded values were zero at addresses 0x20000014 (GNC), 0x20000048 (sensing), 0x2000007C (communications), and 0x200000B0 (telemetry). These reads show that the corresponding counters had not registered activation-period violations at the time of observation.

7.3 Runtime deadline checks

The deadline-miss counters were read separately to avoid treating activation-period checks and deadline checks as the same test. The values were zero at 0x20000018 (GNC), 0x2000004C (sensing), 0x20000080 (communications), and 0x200000B4 (telemetry). The captured values are recorded in the results table and supported by the deadline-counter screenshot.

7.4 Tick configuration and scheduling-period verification

The SysTick LOAD register at 0xE000E014 was inspected and returned 0x0001193F, corresponding to a reload value of 71,999. With a 72 MHz SysTick clock, this is consistent with a 1 ms tick interval. The task period observations were then checked against the configured 1 kHz FreeRTOS tick: GNC 10 ticks, sensing 20 ticks, communications 50 ticks, and telemetry 100 ticks. The minimum and maximum values matched for each task in the RTOS tick-based readings captured for this run.

7.5 High-resolution timer measurements

Timer2 readings were collected separately for each task to retain a finer-grained view of observed periods and execution intervals. The recorded period-count ranges were 99,255–100,027 for GNC, 199,266–200,027 for sensing, 499,208–500,070 for communications, and 999,281–1,000,372 for telemetry. Their corresponding peak-to-peak period variations were 772, 761, 862, and 707 counter counts. The report keeps these as raw counts because the Timer2 input frequency was not independently verified.

Execution-count ranges recorded during the same observation were 90–94 for GNC, 0–21 for sensing, 7–12 for communications, and 6–10 for telemetry. These are instrumented observations, not an independently validated pure workload WCET. RTOS tick-based execution measurements have 1 ms resolution, so a zero tick reading must not be interpreted as zero processor execution.

7.6 Evidence capture and traceability

Ten screenshots are embedded in Section 6 and captioned according to the evidence they show: activation counters, deadline counters, SysTick registers, the two pairs of task-period measurements, four per-task Timer2 measurement views, and the build-artifact directory listing. The screenshots are linked to the corresponding result statements so that a reviewer can compare the report tables with the captured Renode console and Windows directory evidence.

7.7 Task completion summary

Based on the supplied evidence, the build artifacts were present, the expected RTOS tick configuration was observed, all four task periods matched their nominal tick counts in the captured run, and all activation-period and deadline-miss counters read zero. High-resolution Timer2 measurements were recorded for all four task loops and retained in raw-count form. These are the completed and evidenced checks for Task 3; they describe the tested emulation run rather than a claim that every possible runtime condition has been proven.

8. Conclusion

The corrected GNC timing test was built and inspected under Renode. The evidence records the expected RTOS periods for all four task loops, zero observed activation-period violations, and zero observed deadline misses in the captured run. SysTick register inspection supports the configured 1 kHz RTOS tick, and Timer2 readings provide raw high-resolution period and execution measurements for each task. These results document the observed emulation run and the evidence collected for Task 3.

9. Submission Checklist

- Corrected source file included in project directory.
- Corresponding ELF executable included in project directory.
- Renode .resc launch script included in project directory.
- Screenshots for activation counters, deadline counters, SysTick, task periods, Timer2 measurements, and build files included in this report.